FINETUNING WITH UNSLOTH

What is Unsloth?

Unsloth is an open-source project for LLM fine-tuning.It helps you to run the end-to-end pipeline
Unsloth allows you to:

1. Load the model
2. Apply quantization (4-bit, 8-bit, or 16-bit)
3. Train the model
4. Run the inference
5. Evaluate performance
6. Save the trained model
7. Export the model

All of this 2× faster and with up to 70% less GPU memory(VRAM) usage.

Inference Support

Models trained with Unsloth can be directly used with popular inference engines means we can export the trained model in a format so that other tools can use

Unsloth lets you export your model to:

- llama.cpp(gguf)
- Ollama
- vLLM
- SGLang
- Hugging Face Hub
- Open WebUI

These are NOT training tools.
These are inference / deployment tools.

Model Support in Unsloth

Unsloth supports almost all model types:

- Text-to-Text (text generation, chat models)
- Multimodal models (text + image + audio)
- Image-to-Text (OCR, vision-language models)
- Text-to-Speech (TTS)
- Speech-to-Text (STT)
- Vision-language models
- Classical language models like BERT

```
# Hugging Face baseline (NO Unsloth)
# model_name = "TinyLlama/TinyLlama-1.1B-Chat-v1.0"
# HF Transformers
# bitsandbytes 4-bit
# PEFT LoRA
# Standard PyTorch kernels

# Unsloth benchmark
# model_name = "unsloth/tinyllama-bnb-4bit"
# Same base TinyLlama
# But:
# Pre-quantized
# Triton kernels
# Manual autograd
# Auto packing
```

Training types supported

- Full Fine-Tuning
- LoRA
- QLoRA (4-bit and 8-bit)
- 16-bit LoRA
- FP8 Training (supported on compatible hardware)
- Reinforcement Learning (RL) Support

Unsloth has industry-leading reinforcement learning support, including:

- Reward Modeling
- KTO
- PPO
- GRPO
- GSPO
- DPO
- ORPO

Why Unsloth is Special for RL

- Unsloth can use up to 80% less GPU memory for RL workloads
- This makes advanced reasoning and alignment training possible on much smaller GPUs
- Many RL setups that normally require large servers can run on single GPUs or Colab

Goal of Unsloth or Unsloth's Main Claim?

- 2× to 3× faster training
- 50% to 80% less GPU memory (VRAM) usage
- Ability to train large models even on free GPUs like Colab or Kaggle

What Does This Mean in Practice?
If a task normally takes:

- 10 hours and 40 GB VRAM using standard Hugging Face training

With Unsloth, the same task can take:

- ~5 hours and 12–16 GB VRAM

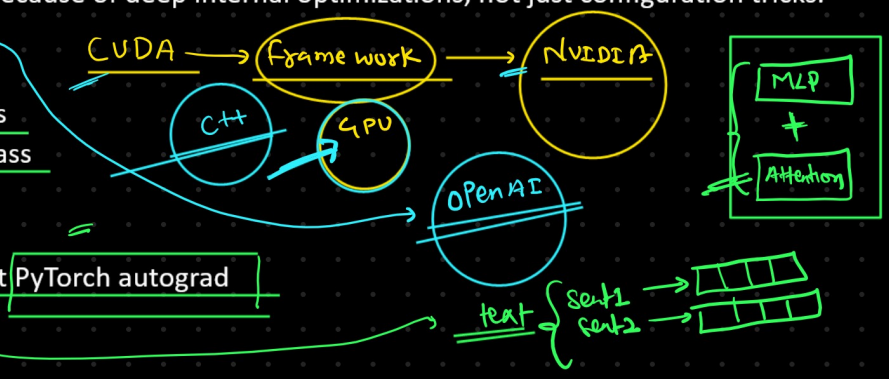
Example: LLaMA 3.1 (8B)

- If Hugging Face needs 24 GB VRAM, Unsloth needs only ~7–8 GB
- If Hugging Face takes 2 hours, Unsloth finishes in ~1 hour

Why is Unsloth so fast and memory efficient?

Unsloth is fast and memory-efficient because of deep internal optimizations, not just configuration tricks.
What Unsloth does internally:

- Custom CUDA & Triton kernels
- Fused attention and MLP operations
- Optimized forward and backward pass
- Smart gradient checkpointing
- Flash-Attention compatibility
- Manual backpropagation engine not PyTorch autograd
- Automatic sequence packing



so there is Exact math (no approximation methods), what is the result of these optimizations?

- Accuracy loss

Practically zero or negligible (no approximation-based degradation)

- Training speed

2× to 3× faster compared to standard Hugging Face training

- GPU memory usage

50%–80% less VRAM

CUDA NVIDIA's low-level C/C++ Libraries
CUDA
Runs on GPU
Massive parallel
Crazy fast
Low-level control

Triton Python-like GPU language
Triton is a Python-based DSL that compiles into highly optimized CUDA kernels.

TOP KEY FEATURES TO HIGHLIGHT

1. Long Context Training

This is Unsloth's biggest shock factor
20K–300K+ tokens training

Same in normal HF training will give OOM error
max limit up to 28K token

Example: LLaMA 3.1 (8B)
Approximate maximum context lengths with Unsloth:

GPU Memory	Max Context (Approx.)
8 GB	~3,000 tokens
12 GB	~21,000 tokens
16 GB	~40,000 tokens
24 GB	~78,000 tokens
80 GB	Up to ~340,000 tokens

These numbers are possible due to Unsloth's memory-efficient kernels, smart checkpointing, and RoPE scaling.

Comparison with Standard Hugging Face Training
Using the same LLaMA 3.1 (8B) model:

- On 12 GB GPU → Out of Memory (OOM)
- On 80 GB GPU → Only ~28,000 tokens supported

2. Massive VRAM Reduction (50–80%)
3. Reinforcement Learning (GRPO, GSPO) at Low VRAM
4. Custom Cuda and Triton Kernels
5. End-to-End Pipeline + Export Freedom
6. Exact Math (No Approximation)
7. Works on Free GPUs (Colab / Kaggle)
8. GPU Support
9. Operating System Support

(PyTorch, Tensorflow, JAX)python based, RUST

Transformer

↓

Hugging Face (Transformers + PEFT + TRL)

↓

Unsloth (Optimized Engine)

↓

CUDA / Triton Kernels

↓

GPU

Unsloth is built on top of Hugging Face Transformers, PEFT, and TRL, but it enhances them with low-level GPU optimizations to deliver 2–3× faster training and 50–80% better memory efficiency.

What solutions can we build in the future using Unsloth?
Text-to-Text: Mistral, deepseek, qwen, gemma, phi, llama
Multimodal:
- Text-to-Speech (TTS): whisper, orpheus
- Speech-to-Text (SST) : whisper large
- Vision- qwen3vl

GRPO (Reasoning RL)
DPO / ORPO (Alignment)